DO NOW — RECALL

# Lesson 080 · Insertion Sort: Grow the Sorted Prefix

Name: _____ Class: _____ Date: _____

**1. What gets fixed after a bubble pass?**

**2. What does insertion-sort key mean?**

**3. What does sorted prefix mean?**

**4. BST root10 with right child15: insert key12. Show comparisons and position.**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Largest remaining value at end.

### 2. Model answer

Item currently being placed.

### 3. Model answer

Initial already-ordered portion.

### 4. Model answer

12>10 so go right;12<15 so insert as left child of15.

### By the end, you can…

• The left portion is kept sorted. Read the next item as key and compare backwards with earlier elements.

• Shift larger elements one position to the right, then insert key in the vacant position.

• Worst $O(n^2)$, already-sorted best O(n), auxiliary O(1); especially useful on small/nearly sorted arrays.

### Vocabulary — know these terms

insertion sort — Insert key into ordered prefix.

key — Value currently being inserted.

shift — Move existing value to make room.

sorted prefix — Already-sorted left section.

NEW LEARNING

# Read, annotate and explain

### Understand

The left portion is kept sorted. Read the next item as key and compare backwards with earlier elements.

### Trace

Shift larger elements one position to the right, then insert key in the vacant position.

### Explain

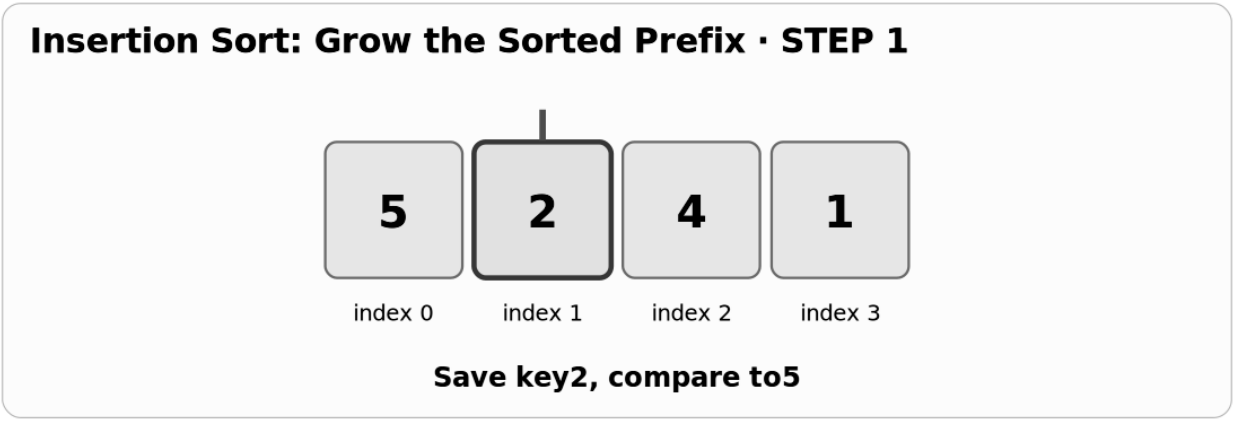
Worst $O(n^2)$, already-sorted best $O(n)$, auxiliary $O(1)$; especially useful on small/nearly sorted arrays.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

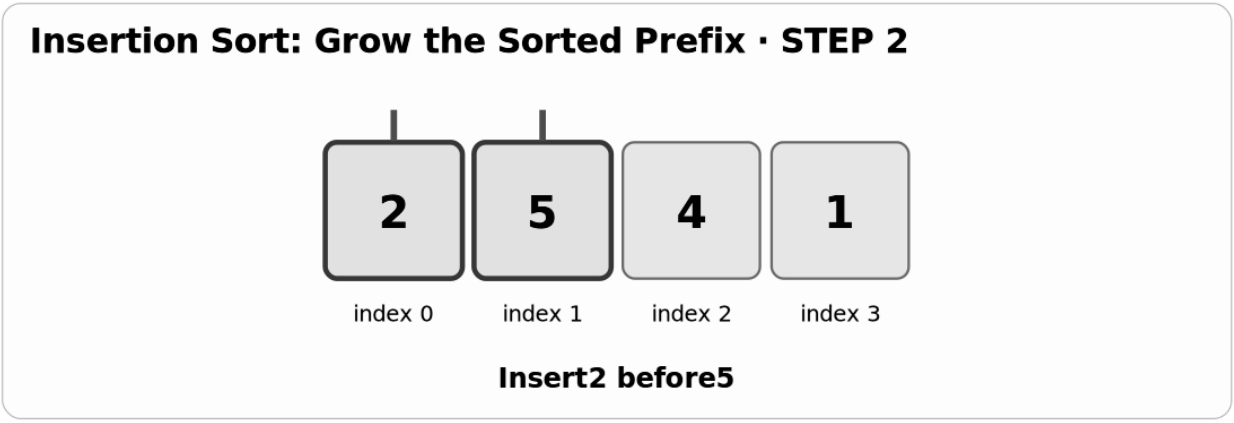
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Insertion Sort: Grow the Sorted Prefix · STEP 1; array 5, 2, 4, 1; Save key2, compare to5

### New Learning

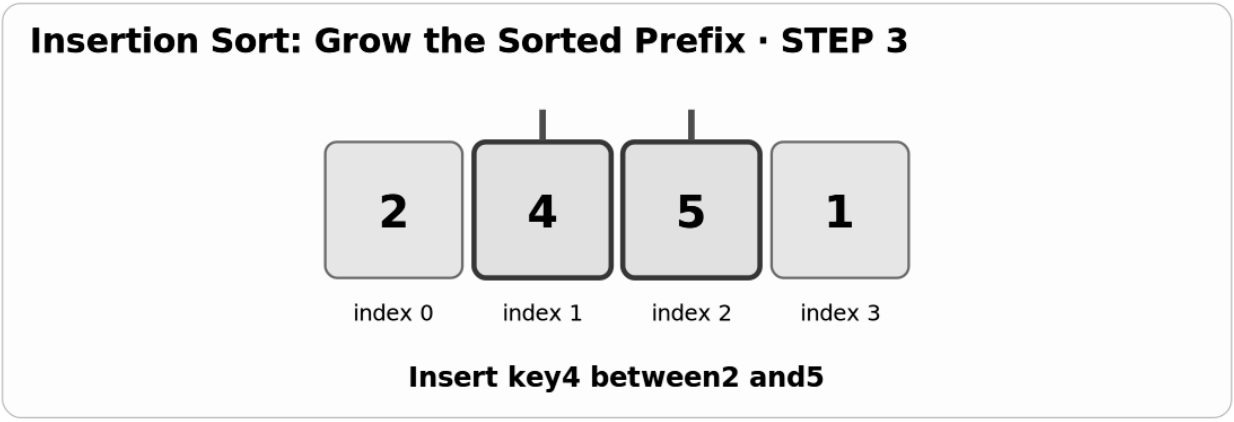



Insertion Sort: Grow the Sorted Prefix · STEP 2; array 2, 5, 4, 1; Insert2 before5

# New Learning · original lesson diagrams

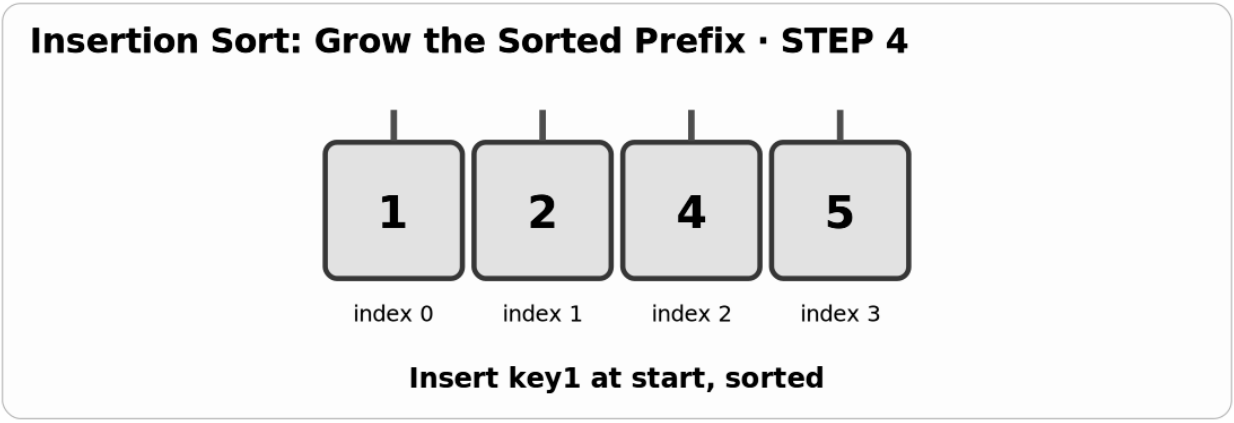
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Insertion Sort: Grow the Sorted Prefix · STEP 3; array 2, 4, 5, 1; Insert key4 between2 and5

### New Learning




Insertion Sort: Grow the Sorted Prefix · STEP 4; array 1, 2, 4, 5; Insert key1 at start, sorted

I DO — WORKED EXAMPLE

# Follow the worked method

### Insertion-sort [4,1,3]; give states after each key.

### Check the working

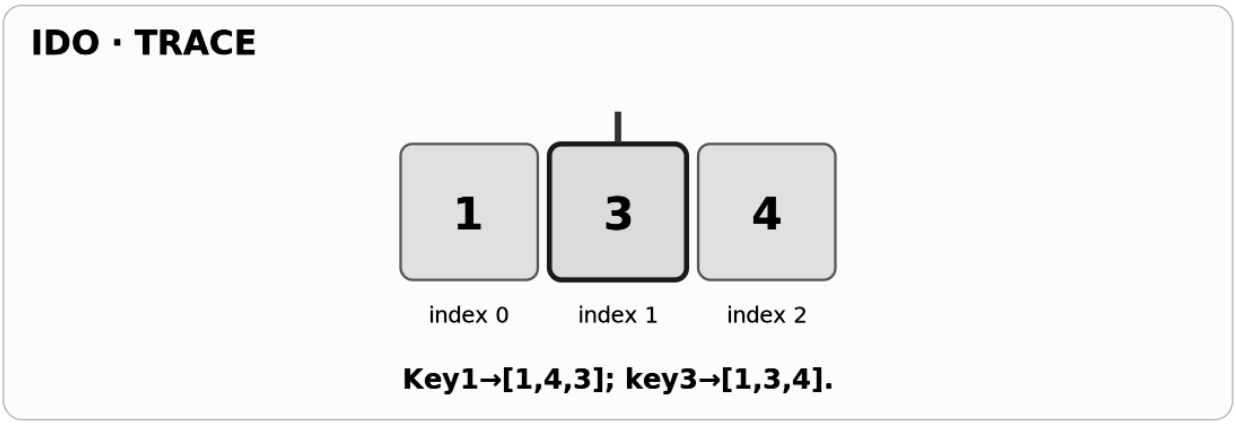
Key1→[1,4,3]; key3→[1,3,4].

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Insertion-sort [4,1,3]; give states after each key.

Key1→[1,4,3]; key3→[1,3,4].




IDO · TRACE; array 1, 3, 4; Key1→[1,4,3]; key3→[1,3,4].

**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

Insert key2 in sorted prefix [3,5] of [3,5,2].
Source code / input:
Shift5 and3 right then write2 at index0, giving [2,3,5].

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Give arrays after first two key insertions into [5,2,4,1]. [6]

### Question 2

Complete insertion of final key1 into [2,4,5,1], showing working. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 3

Compare insertion and bubble sort method/time on already sorted input. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**Insertion sort always requires a separate second array to store sorted values.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

**GREEN PEN — Correct and explain**

Choose one error. Write the correct method and explain precisely why your first response was wrong.

**WE DO — CHECK AND IMPROVE**

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

### Guided 1 — CHECK

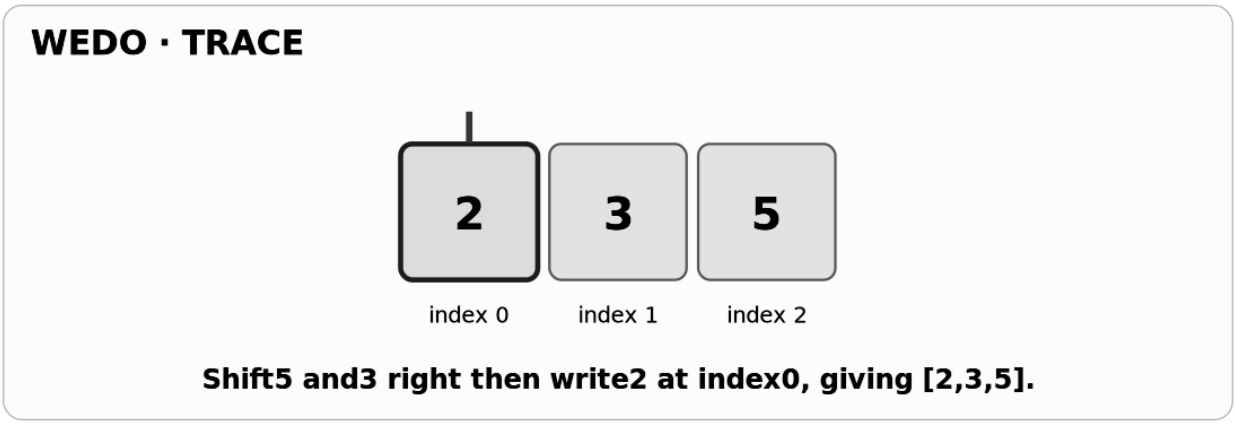
Check the working
Shift5 and3 right then write2 at index0, giving [2,3,5].

# We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Insert key2 in sorted prefix [3,5] of [3,5,2].

Shift5 and3 right then write2 at index0, giving [2,3,5].




WEDO · TRACE; array 2, 3, 5; Shift5 and3 right then write2 at index0, giving [2,3,5].